Product Return Request - AWS Deployment Test Cases

MIS 3371 | Section 26252 | Group 3

Item	Value
Deployed frontend URL	https://staging.d1sg7ta5nv9m14.amplifyapp.com
API Gateway endpoint	https://iqpj8rd7ff.execute-api.us-east-1.amazonaws.com/api/returns (POST)
Lambda function	MIS3371-ReturnRequest (Node.js, execution role: LabRole)
Test dates assume	Today's date = **2026-10-31**. If testing on a different day, shift every Purchase Date by the same number of days. The backend uses the UTC date, which becomes the next calendar day at about 7 pm Central - after that time, treat tomorrow as 'today' when choosing dates.

Rules under test

BR-1 a return must be requested within 30 days of the purchase date (day 30 is allowed, day 31 is not). **BR-2** final-sale and clearance items cannot be returned (flagged SKUs: FS-1001, FS-1002, CLR-2001, CLR-2002). **BR-3** the quantity returned cannot exceed the original quantity purchased. **BR-4** a defective item valued over $200 requires supervisor approval (exactly $200 does not). **Input validation** (in Lambda): required fields, a real calendar date that is not in the future, whole-number quantities of at least 1, non-negative item value, allowed reason / resolution / return method, and a replacement SKU when Replacement is chosen.

Part 1 - Test through the deployed frontend

Open the frontend URL and enter the values below, then press Submit. Unless a row says otherwise, use Customer ID **C-1001**, Order Number **ORD-5001**, SKU **SKU-1001**, Preferred Resolution **Refund**, Return Method **Mail**. Every submission is sent to Lambda through API Gateway; the result panel under the Submit button shows the backend's response (a heading, then the message).

#	Type	SKU	Returned / Original	Purchase Date	Reason	Item ($)	Resolution	Expected result (heading: message)
1	Valid	SKU-1001	1 / 2	2026-10-21	No Longer Needed	50	Refund	**Approved:** Approved.
2	Valid	SKU-1001	1 / 1	2026-10-21	Wrong Item	80	Replacement, Replacement SKU **SKU-5500**	**Approved:** Approved.
3	Business rule (BR-4)	SKU-1001	1 / 1	2026-10-14	Defective	250	Refund	**Approved - pending supervisor review:** Approved, pending supervisor review (defective item over $200).
4	Invalid (BR-1)	SKU-1001	1 / 1	2026-09-16	Wrong Item	40	Refund	**Not approved:** Return request is outside the 30-day return window.
5	Invalid (BR-2)	**FS-1001**	1 / 1	2026-10-21	No Longer Needed	50	Refund	**Not approved:** Item FS-1001 is final-sale and is not eligible for return.
6	Invalid (BR-2)	**CLR-2002**	1 / 1	2026-10-21	No Longer Needed	50	Refund	**Not approved:** Item CLR-2002 is clearance and is not eligible for return.
7	Invalid (BR-3)	SKU-1001	3 / 2	2026-10-23	Other	60	Refund	**Not approved:** Return quantity exceeds the original quantity purchased.
8	Invalid input	SKU-1001	1 / 2	**2026-11-30** (future)	No Longer Needed	50	Refund	**Not approved:** Purchase date cannot be in the future.

#	Type	SKU	Returned / Original	Purchase Date	Reason	Item ($)	Resolution	Expected result (heading: message)
9	Boundary (30 days)	SKU-1001	1 / 1	2026-10-01	Other	75	Refund	**Approved:** Approved. (Exactly 30 days is still inside the window.)
10	Boundary (31 days)	SKU-1001	1 / 1	2026-09-30	Other	75	Refund	**Not approved:** Return request is outside the 30-day return window.
11	Boundary ($200)	SKU-1001	1 / 1	2026-10-28	Defective	**200.00**	Refund	**Approved:** Approved. (Exactly $200 does not need supervisor review.)
12	Boundary ($200.01)	SKU-1001	1 / 1	2026-10-28	Defective	**200.01**	Refund	**Approved - pending supervisor review:** Approved, pending supervisor review (defective item over $200).
13	Boundary (quantity)	SKU-1001	**2 / 2**	2026-10-21	No Longer Needed	50	Refund	**Approved:** Approved. (Returning the full original quantity is allowed.)

Part 2 - Backend validation through the API

The browser's built-in checks stop empty required fields and quantities below 1 before they are sent, so these cases are sent straight to the API (curl, Postman, or the Lambda console test tab). Send the base body below with **only the listed change**. A rejected request returns HTTP **400**; an approved request returns HTTP **200**.

```
curl -X POST https://iqpj8rd7ff.execute-api.us-east-1.amazonaws.com/api/returns -H "Content-Type: application/json" -d 'BODY'
```

Base body: `{"customerId":"C-1001","orderNumber":"ORD-5001","sku":"SKU-1001","purchaseDate":"2026-10-21","originalQuantity":2,"quantity":1,"itemTotal":50,"reasonCode":"No Longer Needed","preferredResolution":"Refund","returnMethod":"Mail","replacementSku":""}`

#	Scenario	Change to the base body	HTTP	Expected message
A1	Quantity of 0	"quantity":0	400	Quantity to return must be a whole number of at least 1.
A2	Replacement with no replacement SKU	"preferredResolution":"Replacement" (leave replacementSku empty)	400	Replacement SKU is required when requesting a replacement.
A3	Missing Customer ID	"customerId":""	400	Customer ID is required.
A4	Unknown reason code	"reasonCode":"Changed my mind"	400	Reason for return must be one of: Defective, Wrong Item, No Longer Needed, Other.
A5	Negative item value	"itemTotal":-5	400	Item value must be a number of 0 or more.
A6	Purchase date is not a real date	"purchaseDate":"2026-02-31"	400	Purchase date must be a valid date (YYYY-MM-DD).
A7	Body is not valid JSON	send `{oops` as the body	400	Request body must be valid JSON.

Notes

Expected outcomes match the backend test suite (`backend/returnRules.test.js`, 20 of 20 passing) and the Lambda test events in the `backend` folder. All business rules and input validation are enforced by the backend; the frontend only shows live hints, so every result in Part 1 comes from Lambda through API Gateway. BR-2 uses a small list of flagged SKUs as a stand-in for a product catalog.